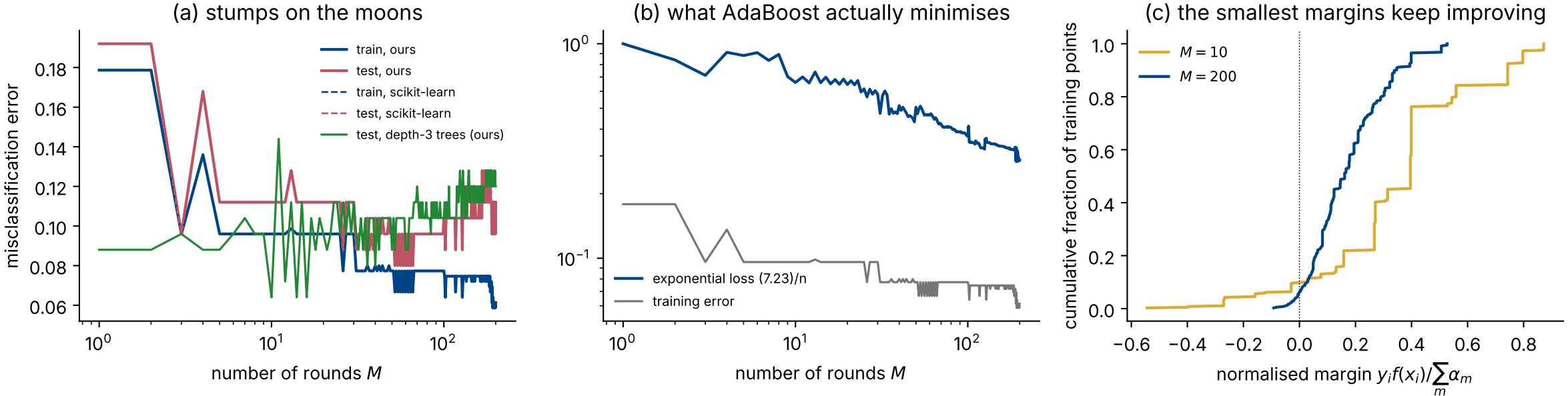
(a) stumps on the moons
misclassification error
0.18
0.16
0.14
0.12
0.10
0.08
0.06
train, ours
test, ours
train, scikit-learn
test, scikit-learn
test, depth-3 trees (ours)
$10^0$
$10^1$
$10^2$
number of rounds $M$
(b) what AdaBoost actually minimises
$10^0$
$10^{-1}$
exponential loss (7.23)/n
training error
$10^0$
$10^1$
$10^2$
number of rounds $M$
(c) the smallest margins keep improving
cumulative fraction of training points
1.0
0.8
0.6
0.4
0.2
0.0
$M = 10$
$M = 200$
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
normalised margin $y_i f(x_i) / \sum_m \alpha_m$